Fiscal Year:	FY 2009	**Task Last Updated:**	FY 03/26/2009
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Targeting specific bone cell signaling pathways and immunosuppression in microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	01/01/2006	**End Date:**	12/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Pellis, Neal	**Contact Phone:**	
Contact Email:	neal.r.pellis@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2008 per PI (12/07) NOTE: Received NCE to 12/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ06HA40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The utility of the above project is apparent, in view of the high level of importance placed by NASA on understanding and developing countermeasures to the effects of microgravity exposure and consequent unloading-induced bone loss and immune suppression in astronauts. Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the skeletal system and the potential use of pharmacological countermeasures such as statins and bisphosphonates. The signaling experiments would delineate target molecules which could be manipulated to maintain equilibrium even in unloaded conditions. Cellular and tissue level changes occur much earlier than whole organ changes, which can be addressed in both these projects, thereby providing a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight perspective. Both these projects will involve travel to national and international NASA related and basic science related symposia and collaborative experiments by the PI. Task Significance: The project will provide a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight perspective.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the immune and skeletal system and the potential use of nutritional and pharmacological countermeasures such as isoflavones, nucleotides,statins and bisphosphonates.
Task Progress:	1. Experiments with gene arrays indicated that some aberrant responses presented themselves in the modeled microgravity scenario in human lymphocytes. The aberrant responses, which included the up-regulation of a detrimental cardiovascular mediator, resulted in another pilot project and accumulation of preliminary data. Collaborations with the V.A. were set up to perform human studies. JSC was also approached for blood samples from human subjects and human protocols were generated and approved. It is evident that Placental induced growth factor (PIGf) levels by ELISA in normal human peripheral blood lymphocytes increases significantly (upto 400% in all donors except donor 3 (300%)) upon exposure to microgravity exposure in vitro. This corroborates previous gene array studies. The up-regulation of PIGf in human lymphocytes upon modeled microgravity exposure after 72 hours indicates that stressful environments such as microgravity can cause elevation of angiogenic factors such as Placental induced growth factor (PIGf) possibly leading to detrimental effects on the cardiovascular system. This has lead to work mentioned in specific aim two of the project, on cardiovascular patients at the V.A. Furthermore recent results indicate the increased expression of the interferon group of genes which are implicated as causative facotrs in autoimmune diseases like rheumatoid arthritis and lupus. 2. Isoflavones exhibited beneficial effects on osteoblasts cultured in modeled microgravity. Preliminary experiments also indicated that the 'Wnt" signaling pathways were sensitive in osteoblasts cultured in modled microgravity. 3. Both the above fulfill the main objectives of the proposal. A collaborative project with Dr. Russomano from IUPCRS , Brazil has resulted in a publication on "immune suppression in modeled microgravity". Altogether many presentations and publications have resulted. Awards: Faculty-Best Presentation Award, Texas Southern University, Research Week, April 2006. Professional Presentations: The preliminary “in vitro” work was written up and sent to the International Academy for Aerospace Medicine for consideration at their annual symposium at Bangkok, Thailand in September 2008. Dr. Sundaresan’s collaborator presented the PlGf data as an oral talk on behalf of Dr. Sundaresan. Title--PlGf as a predictor of cardiovascular dysfunction in microgravity. The protein data was submitted to the NASA, 2009 Human Research Program, investigators' working group meeting and was accepted as an oral presentation in the Cardiovascular Risk session scheduled for February 2, 2009. Title-“Cardiovascular Biomarker expression in microgravity”. The talk at the International Gravitational Space Physiology meeting took place June 23rd in Angers, France. Title-“Validation of Immunological parameters in a 3D clinostat first fabricated in Latin America as a model of simulated microgravity”. Oral presentation at the NAFP symposium in Baltimore, MD, July 2008. Title --“Immune Suppression in Microgravity”. A Sundaresan : Invited speaker : IUPCRS Microgravity laboratory, Porto Alegre, Brazil-“Immune suppression in microgravity” lecture series (four), Nov 22-27, 2008. A Sundaresan : Invited speaker: “Genetic adaptation in microgravity”-University of Loughborough, UK, Dec 13-18, 2008. A. Sundaresan and N.R. Pellis: Upstream targets in Human lymphocytes in modeled microgravity: Manuscript in preparation for “Journal of Cellular Biochemistry”, 2009. Student presentations-Jamail Plumber and Alamelu Sundaresan Historically Black College and University–University Programs (HBCU-UP) Conference. “Analyzing Human Umbilical Vein Endothelium Cells for Gene Expression Deviations Due to Model Microgravity”- Oral Presentation 2008. Texas Academy of Science. “Evaluation of Gas Chromatography-Differential Mobility Spectrometry for Measurement of Air Containments in Spacecrafts”- Poster Presentation

	Texas Academy of Science. “Regulation of Adaptation Genes in Analog Microgravity” – Poster Presentation, 2006.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Articles in Peer-reviewed Journals	Sundaresan A, Pellis NR. "Cellular and genetic adaptation in low-gravity environments." Annals of the NY Academy of Science. 2009 Apr;1161:135-46. http://dx.doi.org/10.1111/j.1749-6632.2009.04085.x ; PubMed PMID: 19426312 , Apr-2009
Articles in Peer-reviewed Journals	Martinelli LK, Russomano T, Dos Santos MA, Falcao FP, Bauer ME, Machado A, Sundaresan A. "Effect of microgravity on immune cell viability and proliferation: simulation using 3-D clinostat." IEEE Engineering in Medicine and Biology. 2009 Jul-Aug;28(4):85-90. http://dx.doi.org/10.1109/MEMB.2009.933572 ; PubMed PMID: 19622430 , Jul-2009
Articles in Peer-reviewed Journals	Sundaresan A, Russomano T, Pellis NR. "Comparison of Changes in Immunological Parameters in Human Lymphocytes in 2d Versus 3d Clinostats-Goal Towards Microgravity Analog Calibration for Future Space Experiments." Journal of Gravitational Physiology, in press March 2009. Expected May 2009. , Mar-2009
Articles in Peer-reviewed Journals	Sundaresan A, Pellis NR. "Upstream targets in Human lymphocytes in modeled microgravity." Manuscript in preparation for Journal of Cellular Biochemistry, March 2009. , Mar-2009
Awards	Sundaresan A. "Faculty-Best Presentation Award, Texas Southern University, Research Week, April 2006." Apr-2006